To measure the resistance of an external resistor, the test leads are connected across the external resistor. The pointer moves on the scale and shows the value of resistor. Different electrical quantities can be measured by inserting the test leads in different jacks for each quantity and range.



Fig. 6.11 **Multi meter**

6.8 Megger

An instrument used for the measurement of high resistance or the insulation resistance is called as megger or insulation tester. The simplified connection diagram of megger is shown in figure 6.12. It has three terminals, namely, the line (L), the earth (E), and the guard.

The moving system consists of three coils known as, the pressure (or control) coil and the current (or deflecting) coil and compensating coil which are mounted rigidly to a pivoted central shaft. The compensating coil is connected in series with the pressure coil in order to provide better scale proportions and to make the arrangement a static. The moving system is free to rotate over a C-shaped iron core. The magnetic field is produced by a pair of permanent magnets for both generator and the moving system.

a. Working principle

Now when the resistance under measurement is connected to the test terminals L and E, the deflecting torque and the controlling torque are produced which oppose each other. At balance, the pointer rests at some intermediate point on the scale. The scale is calibrated in mega ohm and kilo ohm and therefore the value of the resistance under test is indicated by the position of the pointer on the scale.

The required test voltage is generated by a small handle operated permanent magnet DC generator. The generator armature is rotated by hand using the handle at a constant speed which induces the required voltage for testing.

The resistance under test is connected between the terminals L and E and the hand driven generator is rotated at a uniform speed until the pointer shows a stead deflection. In order to maintain constant voltage, a centrifugal clutch is usually provided in the generator drive mechanism which slips at predetermined speed. The test voltages of hand driven megger are 250V, 500V and 1000V.

BEE_ENG_Chapter_06.indd 96 12/27/2019 10:30:45 AM